

SOA Survey

Federated High-Rank Adaptation for Foundation Models

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Abstract

Federated Learning (FL) enables privacy-preserving model training across decentralized clients without sharing raw data, making it attractive for domains such as healthcare, finance, and mobile intelligence. However, extending FL to modern foundation models introduces major communication and computation challenges, since full fine-tuning requires updating and transmitting extremely large parameter sets across repeated federated rounds. Parameter-efficient fine-tuning (PEFT) methods such as LoRA reduce this cost by training only low-rank adapter parameters, but their update matrices remain rank-constrained, which may limit expressiveness when client data are heterogeneous and non-IID.

This survey reviews recent work on Federated Foundation Models, including classical FL optimization methods, personalization strategies, and federated PEFT frameworks. We identify a central gap in current federated LoRA-based approaches: although they improve efficiency, their low-rank updates may be insufficient for capturing diverse client-specific patterns and may worsen aggregation conflicts under statistical heterogeneity. Motivated by Hadamard-based high-rank adaptation methods, including HiRA and BHRA, this report explores FedHiRA and FedBHRA as promising directions for communication-efficient, expressive, and robust federated fine-tuning of foundation models.

1. Problem Definition

Federated Learning (FL) is a distributed training paradigm in which multiple clients collaboratively optimize a shared model without transferring raw local data to a central server. In each round, the server broadcasts a global model to selected clients, clients perform local training using private datasets, and the server aggregates client updates into an improved global model. This approach is motivated by privacy regulations and data ownership constraints, particularly in healthcare, finance, and mobile systems where centralizing data is infeasible.

Recently, foundation models such as large Transformers (LLMs and ViTs) have become dominant due to their strong generalization and transfer capabilities. However, training or fine-tuning these models in an FL setting is significantly more challenging than classical FL. Full fine-tuning requires backpropagation through billions of parameters and repeated transmission of large gradients, making communication and computation prohibitively expensive. Parameter-Efficient Fine-Tuning (PEFT) methods such as Low-Rank Adaptation (LoRA) address this by freezing pretrained weights and learning only small low-rank matrices, drastically reducing the number of trainable parameters. However, standard federated LoRA still suffers from the low-rank bottleneck of the update matrix ($\text{rank}(W) = r$), which restricts the model's ability to capture diverse, client-specific patterns under realistic conditions.

Despite the communication benefits of LoRA, applying it in federated foundation model fine-tuning introduces a fundamental trade-off between efficiency, expressiveness, and robustness. Client datasets in FL are typically non-IID and unbalanced, meaning that each client may learn adapter updates that reflect only a narrow local distribution rather than the global task objective. Since LoRA constrains each update to a low-rank subspace, this restriction may limit the ability of the aggregated global adapter to represent diverse client-specific patterns. At the same time, increasing the rank of LoRA directly increases the number of trainable and communicated parameters, weakening its efficiency advantage. This creates the central problem studied in this report: *how can foundation models be fine-tuned in federated settings while preserving the communication efficiency of PEFT, avoiding the expressiveness limitations of low-rank updates, and remaining robust under heterogeneous client data?*

2. Literature Review

Federated learning research has evolved from classical decentralized optimization methods to modern foundation-model adaptation. Early work established the feasibility of training deep networks without centralizing data using parameter averaging (FedAvg) [1]. However, the advent of large Transformers has shifted focus toward communication-efficient and expressive fine-tuning strategies. This review organizes the literature around the core challenges in federated foundation-model adaptation—statistical heterogeneity, communication overhead, personalization, and the expressiveness limitations of low-rank updates—while highlighting recent Hadamard-based high-rank adaptation techniques (HiRA and BHRA) as promising building blocks for federated PEFT.

2.1. Statistical Heterogeneity and Client Drift

Non-IID client data is a defining property of real-world FL deployments. FedAvg [1] demonstrates empirical robustness to mild non-IID distributions, but later work shows significant degradation under extreme heterogeneity due to *client drift*, where local optimization steps move in directions misaligned with the global objective. FedProx [3] introduces a proximal regularizer that constrains local updates to remain close to the global model, improving stability under heterogeneous data and device participation. Similarly, SCAFFOLD [4] proposes control variates to correct drift by estimating the difference between local and global gradients.

These approaches are effective for smaller models but become less straightforward for foundation models due to high compute costs and memory constraints. In addition, client drift becomes amplified when clients perform multiple local epochs on small datasets, a common scenario in federated instruction tuning. Recent federated PEFT work such as FedIT [6] shows that decentralized instruction tuning is feasible but remains sensitive to heterogeneity and noisy local updates, motivating the need for drift-aware aggregation methods tailored to LoRA updates. Low-rank adapters like LoRA exacerbate these issues because their update matrices

$$\Delta W = BA$$

are strictly rank-constrained

$$\text{rank}(\Delta W) \leq r,$$

limiting their ability to capture diverse client-specific reasoning patterns. This motivates exploration of higher-effective-rank adapters that preserve communication efficiency.

2.2. Communication Efficiency and Parameter-Efficient Fine-Tuning

Communication cost dominates FL rounds for large models. LoRA [2] (and variants such as DoRA) dramatically reduce trainable parameters and communication volume by learning low-rank updates while freezing pretrained weights. Federated PEFT benchmarks (e.g., SLoRA) confirm that such methods narrow the gap to full fine-tuning in NLP tasks while keeping transmission costs low.

Nevertheless, the low-rank inductive bias of LoRA can be overly restrictive for heterogeneous client data. Hadamard High-Rank Adaptation (HiRA) addresses this by formulating the update as $\Delta W = W_0 \odot (BA)$, where W_0 is the frozen pretrained weight matrix. By the Hadamard rank inequality $\text{rank}(O_1 \odot O_2) \leq \text{rank}(O_1) \times \text{rank}(O_2)$, HiRA can achieve effective ranks approaching $\text{rank}(W_0) \cdot r$ while using the same number of trainable parameters as LoRA. Empirical analyses show HiRA produces substantially richer singular-value spectra and higher stable rank than LoRA, yielding consistent gains on commonsense reasoning, dialogue, and mathematical tasks.

Blockwise Hadamard High-Rank Adaptation (BHRA) further refines this idea by partitioning each weight matrix into a $b \times b$ grid and applying HiRA-style modulation independently per block: $\Delta W_{ij} = W_{0,ij} \odot (B_{ij}A_{ij})$. This design preserves the total parameter budget $r(m+n)$ while scaling the attainable rank by a factor proportional to the number of blocks $\approx b \cdot r_0 \cdot r$ and enabling localized capacity allocation. BHRA maintains higher stable rank across rank budgets, mitigates the global modulation collapse observed in vanilla HiRA, and outperforms strong PEFT baselines (LoRA, DoRA, HiRA, ABBA) on Llama-3.2 (1B/3B), Mistral-7B, and Gemma-2 9B across commonsense and arithmetic benchmarks under matched budgets.

These Hadamard-based methods are particularly attractive for FL: they retain LoRA-level communication and inference costs (after merging) while delivering higher-expressiveness updates that better accommodate client heterogeneity.

2.3. Personalization and Model Heterogeneity

A single global adapter often underperforms for minority or outlier clients. Personalized FL approaches such as Per-FedAvg [5] treat the global model as a meta-initialization. pFedLoRA [8] extends this to heterogeneous backbone architectures via proxy feature extractors.

HiRA and BHRA naturally complement personalization. Their blockwise/localized modulation allows different regions of the weight matrix to specialize according to local data statistics, while the inductive bias from W_0 (or its blocks) provides a stable anchor that reduces overfitting on small client datasets. The richer effective rank also alleviates aggregation bias toward dominant clients, as each client can contribute more diverse directions to the global adapter.

Open Problems (flagged by the field:)

- **Robust convergence under extreme non-IID:** Standard averaging of low-rank adapters suffers from subspace misalignment; higher-rank, spatially-localized updates (as in BHRA) may reduce drift while preserving efficiency.
- **Communication-efficient high-expressiveness adapters for foundation models:** LoRA-style methods trade expressiveness for efficiency; Hadamard high-rank families achieve a superior Pareto front but have not yet been studied in federated settings.
- **Aggregation and personalization with heterogeneous adapters:** Blockwise designs introduce opportunities for similarity-aware or importance-weighted aggregation of per-block factors, as well as client-specific block partitioning.
- **Theoretical understanding of federated high-rank adaptation:** Bounds on convergence, effective rank under heterogeneity, and generalization for Hadamard-based updates in FL remain open.

3. Research Gap and Proposed Direction

The literature review establishes that Parameter-Efficient Fine-Tuning (PEFT) methods, particularly LoRA, render federated fine-tuning of foundation models (LLMs and Vision Transformers) practically feasible by drastically reducing communication volume and per-client compute [2]. Federated variants such as FedIT, SLoRA, and pFedLoRA demonstrate the viability of adapter-based collaborative instruction tuning [6, 7, 8]. Classical drift-mitigation techniques such as FedProx and SCAFFOLD further improve stability under moderate heterogeneity [3, 4]. Federated Averaging (FedAvg) remains the foundational optimization framework underlying most decentralized training approaches [1].

However, a critical gap remains when scaling these approaches to real-world foundational federated learning under severe statistical heterogeneity, non-IID client distributions, small local datasets, and intermittent participation. Current federated LoRA frameworks are fundamentally constrained by the low-rank bottleneck of the update matrix $\Delta W = BA$ ($\text{rank}(\Delta W) \leq r$). This restriction limits each client’s ability to capture diverse, task-critical local patterns, resulting in:

- **Subspace misalignment during aggregation:** Client updates often span conflicting low-dimensional subspaces, amplifying client drift and destabilizing the global model.
- **Insufficient expressiveness:** Low-rank updates struggle to model heterogeneous reasoning transformations across clients.
- **Poor personalization and minority generalization:** The global adapter cannot effectively recombine the narrow directions contributed by individual clients.

Recent Hadamard-based high-rank adaptation methods—HiRA ($\Delta W = W_0 \odot (BA)$) and especially its blockwise extension BHRA—offer a compelling remedy. By leveraging the Hadamard product inequality, these techniques achieve substantially higher effective rank (approaching $r_0 \cdot r$ for HiRA and $b \cdot r_0 \cdot r$ for BHRA) while preserving identical trainable parameter counts and communication costs as LoRA.

BHRA’s localized blockwise modulation further enables spatially controllable capacity allocation and maintains richer singular-value spectra across layers and rank budgets, as validated on Llama-3.2, Mistral-7B, and Gemma-2 models.

3.1. Proposed Direction

We advocate extending these high-rank Hadamard adapters into the federated regime through **FedHiRA** and **FedBHRA**. The core idea is to replace (or augment) low-rank LoRA adapters with blockwise Hadamard high-rank adapters at each client, then develop tailored federated aggregation strategies that operate on the low-rank factors B_{ij} , A_{ij} while respecting the inductive bias encoded in the frozen weights W_0 .

3.2. Contribution Statement

While prior work has enabled communication-efficient federated PEFT, it has not addressed the expressiveness–stability trade-off imposed by low-rank updates under realistic client heterogeneity. We propose FedHiRA and FedBHRA—the first federated instantiations of Hadamard high-rank adaptation—together with drift-aware, similarity-weighted, and block-level aggregation mechanisms. This direction aims to deliver more expressive, robust, and communication-efficient fine-tuning of foundation models in decentralized environments.

3.3. Research Questions

1. **Expressiveness and Heterogeneity:** To what extent do FedHiRA and FedBHRA improve convergence speed, final accuracy, and robustness to severe non-IID distributions compared to federated LoRA baselines under matched communication budgets?
2. **Aggregation Design:** Can blockwise or similarity-aware aggregation rules (e.g., weighted averaging of per-block low-rank factors guided by local stable-rank statistics or client embedding similarity) effectively mitigate client drift while preserving the rank amplification benefits of BHRA?
3. **Theoretical and Practical Trade-offs:** What are the convergence guarantees and generalization properties of federated high-rank Hadamard adaptation? How do choices of block size b , per-block rank $\bar{r}$, and merging strategies impact communication efficiency, personalization, and scalability to large foundation models in cross-device and cross-silo settings?

Subsequent sections will formalize the FedBHRA algorithm, analyze its theoretical properties, and outline an empirical evaluation plan across heterogeneous federated benchmarks for instruction tuning and multimodal foundation models. This research trajectory directly addresses the identified gap by marrying the communication efficiency of PEFT with the representational power of localized high-rank adaptation.

4. Benchmark, Evaluation, and Data Landscape

Datasets. Federated fine-tuning of foundation models typically leverages publicly available instruction-tuning and reasoning datasets that can be partitioned across clients to simulate non-IID distributions (via Dirichlet sampling or task-based splits). The table below summarizes representative datasets used in recent federated LLM studies and suitable for evaluating FedHiRA/FedBHRA.

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Evaluation Protocol. For ****FedHiRA**** and ****FedBHRA****, evaluation follows established practices in federated PEFT while emphasizing the advantages of high effective-rank adaptation. Primary metrics include:

- ****Global performance****: Macro-averaged accuracy on commonsense reasoning (8-task suite) and exact-match (EM) / normalized accuracy on arith-

Table 1: Key datasets for Federated Foundation Model Adaptation.

Dataset	Size	Modality	Notes
Dolly-15k	15k	Text	Commonly partitioned for non-IID simulations; diverse instruction categories.
Natural Instructions (NI)	In- ~738 tasks	Text	Task-per-client splits create strong feature skew; used in FedIT benchmarks.
Commonsense Reasoning Suite	170k+	Reasoning	Includes BoolQ, PIQA, SIQA, HellaSwag, ARC, and OBQA.
MetaMathQA	50k+	Math	Used for arithmetic reasoning evaluation on GSM8K and MATH.
FLAN Alpaca-style	/ 50k–100k	Text	Supports Dirichlet partitioning ($\alpha \in [0.1, 5.0]$).

metric benchmarks (GSM8K, MATH). - **Federated-specific metrics**: Worst-client accuracy, client accuracy variance, and personalization gap (local vs. global adapter performance). - **Efficiency metrics**: Total communication volume (bytes per round, proportional to rank r), number of communication rounds to target accuracy, and post-merging inference latency (identical to base model after folding adapters). - **Expressiveness diagnostics**: Stable rank and singular-value spectrum of aggregated updates to quantify the benefit of blockwise Hadamard modulation over standard LoRA under heterogeneity.

All experiments should report results averaged over multiple random seeds and varying heterogeneity levels (α) to demonstrate robustness.

Compute Context. Recent federated foundation model studies operate on GPU clusters (A100/H100 or V100) for server-side aggregation and multi-client simulation. Thanks to the PEFT nature of HiRA/BHRA, individual clients can perform local

training on modest hardware (single A100 or even RTX 4090 for 1B–7B models) with communication budgets identical to LoRA ($r(m+n)$ parameters per adapter). After training, adapters are merged into the frozen weights, yielding zero additional inference overhead. This setup makes FedBHRA particularly attractive for cross-device and cross-silo deployments where communication and per-client memory are the primary constraints.

Experimental Design Considerations

A rigorous evaluation of **FedHiRA** and **FedBHRA** must demonstrate that the high effective-rank and localized modulation properties of Hadamard adapters translate into tangible gains in federated settings—particularly under severe statistical heterogeneity—while maintaining LoRA-level communication efficiency.

Baselines. Experiments should compare against both classical FL optimizers and state-of-the-art federated PEFT methods. Essential baselines include:

- **FedAvg** [1], **FedProx** [3], and **SCAFFOLD** [4] applied on top of standard LoRA adapters (naive averaging of low-rank matrices).
- Federated LoRA with simple averaging (the most direct counterpart).
- Strong federated PEFT baselines: **SLoRA** [7] and **pFedLoRA** [8].
- **FedHiRA** (global Hadamard modulation) and **FedBHRA** (blockwise variant) with identical parameter budgets.

Where possible, include full fine-tuning (when feasible on smaller backbones) and centralized HiRA/BHRA as performance upper bounds.

Ablation Studies and Hyperparameters. Ablations should systematically isolate the benefits of high-rank Hadamard adaptation:

- Block size b and per-block rank $\bar{r}$ in FedBHRA (e.g., $b \in \{4, 8, 16\}$ with $b \times \bar{r} \approx r_{\text{total}}$).
- Heterogeneity level via Dirichlet concentration parameter $\alpha \in \{0.1, 0.5, 1.0, 5.0\}$.

- Number of local epochs E and client participation rate (e.g., 10%–50%).
- LoRA/HiRA/BHRA rank budget r (e.g., 8, 16, 32, 64) under fixed communication volume.
- Aggregation strategies: simple averaging vs. drift-aware (SCAFFOLD-style control variates on low-rank factors), similarity-weighted, or stable-rank-based weighting of per-block updates.

Additional ablations should examine the impact of merging strategy (pre-merge vs. post-aggregation) and whether block partitioning is shared globally or adapted per-client.

Evaluation Protocol and Metrics. Following the benchmarks in Section 4, report:

- **Global & Client-level Performance:** Macro accuracy on commonsense reasoning (8-task suite), EM on GSM8K/MATH, worst-client accuracy, and client accuracy variance.
- **Expressiveness:** Stable rank and singular-value distribution of the aggregated ΔW across layers.
- **Efficiency:** Total communication volume (bytes), communication rounds to target accuracy, and wall-clock time.
- **Robustness:** Training stability (loss variance, divergence rate) and personalization gap (local vs. global adapter).

Common Pitfalls to Avoid. Unrealistic IID or centrally-shuffled partitions should be avoided; all data partitioning must respect strict non-IID client splits. Reporting only global average accuracy while ignoring worst-client performance or personalization is insufficient. Validation must use purely local client-held evaluation sets. Finally, all methods must be compared under strictly matched communication budgets and total trainable parameters to ensure fair assessment of the rank-amplification benefits of Fed-HiRA/FedBHRA.

This experimental design, inspired by the evaluation protocols in HiRA [9], BHRA [10], FedIT, and SLoRA, provides a comprehensive and reproducible

framework to validate whether blockwise Hadamard high-rank adaptation delivers superior robustness, expressiveness, and efficiency in realistic federated foundation model scenarios.

5. Annotated Reading List

Every paper you read must appear here, minimum 8–10 entries. Three lines of annotation per paper are mandatory. Do not use a separate .bib file.

Paper 1 - Survey Paper

The authors emphasize that "model drift" caused by non-IID data distributions and the massive cost of parameter exchange remain the dominant hurdles for practical deployment. This work serves as a foundational reference for exploring more efficient aggregation strategies and justifies the need for adaptive approaches

Paper 2 - FedAvg

Introduced the Federated Learning framework and the FederatedAveraging (FedAvg) algorithm, which decentralizes deep network training by aggregating locally-computed SGD updates from mobile devices. FedAvg is robust to non-IID and unbalanced data distributions, reducing required communication rounds by 10–100x compared to synchronized SGD. As the foundational baseline for all decentralized optimization, this paper is the primary building block for our work

Paper 3 - LoRA

Introduced Low-Rank Adaptation (LoRA), which freezes pre-trained model weights and injects trainable rank decomposition matrices into Transformer layers to perform parameter-efficient fine-tuning. For GPT-3 175B, LoRA reduces trainable parameters by 10,000x and GPU memory requirements by 3x. This is the baseline for Federated PEFT.

Paper 4- Shepherd Framework

Introduced Federated Instruction Tuning (FedIT), a framework that applies federated learning to the instruction-tuning phase of Large Language Models (LLMs) to leverage diverse, private, on-device user data. This paper establishes the primary conceptual baseline for decentralized LLM adaptation, providing

a research-ready codebase (Shepherd) that we can use to validate novelty in federated foundation models. The authors acknowledge that existing federated tuning often consumes all local data, which can lead to computational overhead and overfitting on small, noisy edge-side datasets.

Paper 5 - pFedLoRA

Proposed pFedLoRA, a framework for personalized federated learning that enables clients with heterogeneous local model architectures to collaboratively tune a shared, small LoRA adapter. Achieved a 1.35% increase in test accuracy while simultaneously reducing computational overhead by 11.8x and communication costs by 7.4x. It uses a Proxy Homogeneous Feature Extractor to bridge the gap between a client running a 3B model and one running a 7B model.

Paper 6 - Survey 2

Provides a comprehensive survey and functional architecture that traces the evolution from distributed machine learning (DML) to federated learning (FL). The survey notes that synchronous updates (FedAvg) are the standard but suffer from "stragglers". Identifies that while FL inherits parallelization techniques from DML, it fundamentally differs by prohibiting raw data transfer and addressing the unique challenges of non-IID data and legal/privacy constraints.

Paper 7 - sLoRA

Evaluated the performance of standard Parameter-Efficient Fine-Tuning (PEFT) methods in Federated Learning for NLP and proposed SLoRA. SLoRA achieves comparable performance to full-parameter tuning with only 1% parameter density, reducing training time by up to 90%. This provides the SLoRA baseline that we either adopt or outperform in our Foundation Model work

Paper 8 - SCAFFOLD

SCAFFOLD achieves communication savings proportional to the number of local steps E without degrading convergence — something FedAvg cannot guarantee. SCAFFOLD represents the "correct but expensive" approach. If our project targets a cross-silo medical imaging scenario (e.g., 5-10 hospitals), SCAFFOLD's statefulness is fine. If we target cross-device, we need something stateless.

Paper 9 - PerFedAvg

instead of learning one global model that works for everyone, learn a global initialization that each client can quickly fine-tune to their own data. Provides formal convergence guarantees for personalized FL. Shows that personalization and global learning are not conflicting.

Paper 10 - FedProx

FedProx handles systems heterogeneity by allowing variable amounts of local work per client. Shows convergence depends on two quantities: B-local dissimilarity (how different client gradients are from the global gradient) and the number of active devices per round. Limitation: Under extreme non-IID (Dirichlet $\alpha \downarrow 0.1$), FedProx still struggles significantly.

References

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Arcas. Communication-efficient learning of deep networks from decentralized data. *AISTATS*, 2017. *What it does: It introduces the Federated Learning framework and the FederatedAveraging (FedAvg) algorithm to train deep networks on decentralized data. What it contributes: It demonstrates that simple model averaging is robust to non-IID data and reduces communication rounds by 10-100x. Why it is here: Foundational baseline for all decentralized optimization in federated foundation models.*
- [2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. *What it does: Introduces low-rank adapter matrices for parameter-efficient fine-tuning of large models. What it contributes: Dramatically reduces trainable parameters while preserving performance. Why it is here: Core technical baseline for federated PEFT approaches.*
- [3] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith. Federated optimization in heterogeneous networks. *MLSys*, 2020.

- [4] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning. *ICML*, 2020.
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 - [6] J. Zhang et al. Federated instruction tuning of large language models. *Findings of ACL*, 2023.
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 - [9] Q. Huang, T. Ko, Z. Zhuang, L. Tang, and Y. Zhang. HiRA: Parameter-efficient Hadamard high-rank adaptation for large language models. *ICLR*, 2025.
 - [10] F. Yu, J. Hu, and G. Min. Blockwise Hadamard high-rank adaptation for parameter-efficient LLM fine-tuning. *arXiv preprint arXiv:2509.21637*, 2025.
- <https://github.com/tloen/alpaca-lora>
Useful background for understanding how LoRA is implemented in practice but has no FL component.
 - <https://github.com/Lightning-AI/lit-llama>
Useful background for understanding how LoRA is implemented in practice but has no FL component.

6. Preliminary Implementation Resources

5–10 GitHub links that you have actually looked at. For each one, one line on what it is and why it is on your list: is it a baseline you might compare against, a dataset loader, a reference implementation of a method you surveyed? You are not committing to using any of these.

- <https://github.com/mtuann/federated-learning-updated-papers>
List of Papers (highly relevant)
- <https://github.com/lishenghui/awesome-fm-fl>
Contains a survey of relevant repos and papers
- <https://github.com/chenxingqiang/PFLoRA-lib>
Can use this as codebase instead of building from scratch